Homework 6

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Model Specification (25 points)

1. Outcome and link function (10 points): Identify the binary, count, or discrete outcome you are analyzing. Specify an appropriate link function and variance function for your outcome. Justify your choices.

Our outcome is systolic blood pressure which is a continuous response so we will use the identity link specified as: $g(\mu_{ij}) = \mu_{ij} = X_{ij}\beta$ where $E(Y_{ij}) = \mu_{ij}$. The marginal variance will be specified as $Var(Y_{ij}) = \phi$.

2. Mean model (15 points): Write out your marginal mean model. Identify the covariates you are including and describe what each parameter represents. Remember that parameters in marginal models have population-averaged interpretations.

Let Y_{ij} be the systolic blood pressure for subject i at the jth visit. Note that marginal models handle time continuously where it enters as a covariate in X_{ij}. Additionally, since marginal models do not include a random effect, we will only have the population mean trajectory (no subject specific mean trajectory). We can write the marginal mean model as:

$$E(Y_{ij}) = g^{-1}(X_{ij}\beta) = \mu_{ij} = \beta_0 + \beta_1 x_{i,YEAR} + \beta_2 x_{i,SEX}$$

- β_0 denotes the baseline mean SBP for males
- β_1 denotes the change in mean SBP per year
- β_2 denotes the difference in mean SBP between females and males
- $x_{i,YEAR}$ denotes the number of years since baseline for the ith subject
- $x_{i,SEX}$ is a binary indicator for sex of the ith subject

Working Correlation Structure (25 points)

1. Fitting the model (10 points): Fit your marginal model using GEE with at least two different working correlation structures (e.g., independence, exchangeable, AR(1), unstructured). Report the parameter estimates from each.

We will fit our model with three different working correlation structures: exchangeable (compound symmetry), AR(1), and unstructured. Table 1 below gives the parameter estimates, standard errors, Wald test statistics, p-values, and the QIC for each model.

2. Comparing structures (15 points): Compare the results across working correlation structures. How sensitive are your parameter estimates to the choice of working correlation? Select a working correlation structure and justify your choice.

Table 1: Comparison of GEE estimates across working correlation structures

Working Correlation	Parameter	Estimate	SE	Wald Statistic	p-value	QIC
Compound Symmetry	(Intercept)	129.263	0.452	81926.71	<0.0001	4461425
Compound Symmetry	YEAR	0.838	0.028	901.85	<0.0001	4461425
Compound Symmetry	SEX	1.807	0.656	7.59	0.0059	4461425
AR(1)	(Intercept)	129.170	0.451	82210.57	<0.0001	4461551
AR(1)	YEAR	0.838	0.028	901.85	<0.0001	4461551
AR(1)	SEX	1.772	0.659	7.24	0.0071	4461551
Unstructured	(Intercept)	129.263	0.450	82519.48	<0.0001	4461443
Unstructured	YEAR	0.833	0.028	860.22	<0.0001	4461443
Unstructured	SEX	1.803	0.653	7.62	0.0058	4461443

Regardless of the covariance structure, all terms are found to be significant in the model and all estimates are generally very similar to each other. This tells us that our parameter estimates are not very sensitive to the choice of working correlation. This aligns with the fact that parameter estimates are consistent even when the covariance structure is misspecified. When looking at the QIC for each model, we yield the smallest QIC under compound symmetry covariance structure (exchangeable). Because of this, we will choose compound symmetry as our working covariance structure.

Variance Estimation (20 points)

1. Comparison (10 points): For your selected working correlation structure, compare the model-based and sandwich (robust) standard errors. Create a table showing both sets of standard errors.

Table 2: Comparison of model based and robust standard errors for exchangable covariance structure

Working Correlation	Parameter	Estimate	SE	Wald Statistic	p-value	QIC
Compound Symmetry	(Intercept)	129.263	0.452	81926.71	<0.0001	4461425
Compound Symmetry	YEAR	0.838	0.028	901.85	<0.0001	4461425
Compound Symmetry	SEX	1.807	0.656	7.59	0.0059	4461425
AR(1)	(Intercept)	129.170	0.451	82210.57	<0.0001	4461551
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2. Interpretation (10 points): Discuss what the differences (or similarities) between these standard errors suggest. Given your sample size and the robustness properties of GEE, which variance estimator do you consider more appropriate for your analysis, and why?

The model-based standard errors are similar to the robust standard errors. The biggest deviance comes from the intercept where the model based standard error is greater than the robust standard error by approximately 0.09. Both year and sex yield model based standard errors that are very close to the robust standard errors. The similarity between these errors suggests that the selected working covariance structure is a reasonable specification. Despite this, we will still use the robust standard errors to be safer as the model based standard errors rely on the covariance structure being correctly specified.

Hypothesis Testing and Interpretation (20 points)

1. Hypothesis test (10 points): State and test a hypothesis relevant to your research question. Report the test statistic, degrees of freedom, p-value, and interpret the result.

We will test to see if there is a sex effect. Our hypotheses are $H_0 : \beta_2 = 0$ vs $H_1 : \beta_2 \neq 0$. We will look at compound symmetry, sex row of Table 2 to find the relevant statistics to report for this test. This test has 1 degree of freedom and yields a test statistic of 7.59 and a p-value of 0.0059. Since the p-value is < 0.05 we will reject the H_0. We claim that there is a significant difference in the mean SBP between males and females.

2. Interpretation of effects (10 points): Interpret your key parameter estimates in the context of your research question. Be specific about the population-averaged interpretation (e.g., for a binary outcome with logit link, interpret in terms of odds ratios at the population level).

The compound symmetry section of Table 2 presents the parameter estimates discussed below:

- $\beta_0 = 129.263$ represents the population-averaged mean SBP at baseline for males
- $\beta_1 = 0.838$ indicates that, on average, SBP increases by about 0.838 mmHG per year at the population level
- $\beta_2 = 1.807$ indicates that on average, females have an SBP that is 1.807 mmHG greater than males, after adjusting for year

Recall that we have an continuous outcome and used the identity link.

Summary (10 points)

Summarize your findings as you would for a collaborator. Address:

- What model did you fit and what working correlation structure did you use?
- What are the key findings regarding your research question?
- What are the limitations of the marginal model approach for your data?

We fit a marginal model for systolic blood pressure using generalized estimating equations with an identity link, including time and sex as covariates. We specified an exchangeable (compound symmetry) working correlation structure to account for within-subject correlation across repeated measurements.

We found that both time and sex are significantly associated with systolic blood pressure at the population-level. Specifically, at the population level, we found that SBP increases over time and females have higher SBP than males on average.

One limitation of the marginal model is that we are not able to calculate subject-specific mean trajectories since a random effect is not included in our model. Because of this, we are not able to capture individual-level variability in baseline SBP or its change over time. Lastly, the model-based standard errors require that the working covariance structure is correctly specified, however the robust standard errors provide an alternative for when the covariance structure may be misspecified.